

The suggested project topics for students of AIIT.

Prof. (Dr.) Alpana Goel

- 1) Physical Protection system for the facility using AI and Machine learning.**

Dr. Arpita Datta

- 2) Artificial intelligence for Nutritional Element Profiling in Food Samples.**

Nutritional element profiling of food samples is essential for evaluating food quality, safety, and dietary value. Conventional elemental analysis techniques generate large datasets that require careful interpretation. This project aims to apply AI techniques to analyze elemental composition data of food samples and automatically classify or predict their nutritional profiles and also to detect presence of toxic elements if any. AI model development will be based on Basic machine learning algorithms will be applied using programming tools such as Python with libraries such as Scikit-learn/ TensorFlow/Keras

- 3) Python-Based Machine Learning Model for Radiation Hazard Prediction in Environmental Samples.**

Environmental samples such as soil, food, water may contain naturally occurring radioactive isotopes. Monitoring radiation levels in these samples is essential for assessing environmental safety and potential health risks. This project aims to develop a Python-based machine learning model that can analyse radiation measurement data and predict radiation hazard levels in environmental samples.

Ms. Archana Yadav

- 4) Prediction of optimized Combination of shielding materials required for gamma radiations using techniques of AI.**

ANN based predictions performed on MATLAB for this data indicate fair accuracy. The simulated results will be further used for making compound shield for attenuating gamma rays.

- 5) Prediction algorithm using advanced AI tools for localising a radioactive source in a 3D geometry which will be used in baggage scanning applications.**

Preliminary work has been done with ANN, and the prediction accuracy can be improved.

- 6) Simulation of a radioactive contamination and its dispersion in a room as per the ambient temperature, pressure and air circulation conditions.**

Further estimation of doses to individuals due to inhalation of the contaminants

- 7) Simulation of a radioactive plume dispersion in air and transportation in different zones of the ground.**

Estimation of radioactive spillage and doses received by individuals.

Dr. Unnati Gupta

- 1) Determination of isotope activity for building materials using gamma ray spectrometry based on neural network model**

Building materials (cement, bricks, granite, etc.) may contain naturally occurring radionuclides (e.g., ^{226}Ra , ^{232}Th , ^{40}K). Their activity concentration must be measured to assess radiological safety. AI-based models reduce human bias, improve detection in complex spectra, and allow rapid screening of materials for compliance with safety standards.

2) **Determination of base line of radiation alarm systems using gamma monitoring devices.**

Establishing base line ensures confidence in reporting whether a radionuclide is present or below detection limits, critical for environmental monitoring and nuclear safety.

3) **Determination of cross section: finding the probability of a nuclear reaction using AI (Physics informed)**

Nuclear cross section quantifies the probability of a reaction (e.g., neutron capture, fission, scattering) occurring when a particle interacts with a nucleus. Machine learning can interpolate and extrapolate cross section data across energy ranges. Neural networks trained on evaluated nuclear data libraries (ENDF, JENDL) can predict cross sections for isotopes with limited experimental data. AI accelerates nuclear data evaluation, improves reactor simulations, and supports applications in medical isotopes, shielding design, and astrophysics.

4) **Spectrum Analysis of ^{155}Eu for calibration of p-type HPGe detector using**

^{155}Eu (Europium-155) is a radionuclide emitting gamma rays at specific energies (notably around 86 keV and 105 keV).: p-type semiconductor detectors (e.g., HPGe) require precise energy calibration to ensure accurate peak identification. Identify characteristic peaks and assign them to known energies. Automated Peak Detection: AI models can identify ^{155}Eu peaks even in noisy spectra.

Dr. Sudatta Ray

8) **Machine learning assisted Neutron – gamma discrimination for a liquid scintillator detector.**

Neutron–gamma discrimination in a liquid scintillator detector is achieved by analysing differences in pulse shapes of detected signals. Machine learning (ML) models are trained on labelled waveform data to automatically classify neutron and gamma events with higher accuracy. In addition, Physics-Informed Neural Networks (PINNs) incorporate underlying physical principles of scintillation and decay processes into the learning model. This improves robustness and generalization, especially in noisy or low-signal conditions. Such hybrid ML–PINN approaches enhance real-time and reliable particle identification.

9) **Machine learning assisted efficiency profile determination at different energies for liquid scintillator.**

Efficiency profile determination for a liquid scintillator across different energies can be enhanced using machine learning (ML) by learning the mapping between detected signal features and true detection efficiency. Training data may come from calibration experiments or simulations over a range of energies. Physics-Informed Neural Networks (PINNs) further improve this by embedding physical laws such as energy-dependent light yield and detector response into the model. This reduces reliance on large datasets and ensures physically consistent predictions. The ML–PINN approach enables accurate, smooth efficiency curves even in sparse or noisy measurement conditions

10) **Optimization of shielding materials for neutrons using AI.**

Optimization of neutron shielding materials using AI involves training models on simulation and experimental datasets to predict attenuation performance of different material compositions. AI techniques such as neural networks and evolutionary algorithms can efficiently explore large design spaces of material mixtures and thicknesses.

Dr. Abhishek Yadav

1) RADMAP - RADIation MAPping – GPS enabled radiation mapping prototype on Raspberry Pi

A prototype will be developed by integrating a radiation detector such as a GM tube or NaI-based sensor with a GPS module (initially we can go with some other sensors, like temperature, moisture etc). The detector will be interfaced with a microcontroller (e.g., Arduino or Raspberry Pi) to enable real-time acquisition of radiation counts along with corresponding geographical coordinates (GPS). The collected data will be visualized using a map-based interface to generate radiation heatmaps. The system will be tested in field conditions to evaluate its accuracy, reliability, and spatial consistency.

2) NUCLID - AI based Radioisotope identifier program

For the radioisotope identifier using a NaI(Tl)-based RIID, gamma-ray spectra will be acquired using a NaI(Tl) scintillation detector and processed through suitable signal conditioning techniques such as noise reduction, smoothing, and energy calibration. Peak detection algorithms will be implemented to identify characteristic gamma energies, which will then be matched against a reference library of commonly used standard radioisotopes. An identification framework, either neural network or using basic machine learning methods, will be developed and validated by testing with known and unknown sources to assess its accuracy and robustness.

3) SHIELD-X: SHIELDing Optimization for EXtreme Environments

For the optimization of radiation shielding in outer space, the space radiation environment, including galactic cosmic rays and solar particle events, will be modeled using appropriate simulation tools. Various candidate shielding materials such as lunar regolith, polyethylene, and composite structures will be defined and analyzed. Monte Carlo simulations will be performed to study radiation transport and estimate dose deposition within a human-equivalent phantom. A parametric study will be carried out by varying material composition and thickness to evaluate shielding effectiveness, including the impact of secondary radiation. Based on the results, an optimized multi-layer shielding configuration will be proposed to minimize radiation exposure in space environments.

4) AtomMitra

In the digital age, mobile applications provide a powerful medium to spread scientific knowledge. Nuclear science, though widely used in energy, medicine, and industry, is

often misunderstood due to limited awareness. This project aims to develop a simple and user-friendly mobile app to promote understanding of nuclear science among students, general public and up to some extent nuclear professionals. The app will present key concepts in an easy-to-understand format using visuals, short explanations, and interactive quizzes. It will also highlight real-life applications, radiation safety, and recent developments in the field. By making information accessible and engaging, the app will help improve scientific awareness and encourage informed thinking about nuclear science and technology.